

Answers to Odd-Numbered Problems

Chapter 1

Answers to odd-numbered multiple-choice problems

9. (b)
11. (c)
13. (a)
15. (c)
17. (a)

Answers to odd-numbered problems

19. (a) 1.395×10^4 m;
(b) 2.46×10^{-5} kg;
(c) 3.49×10^{-8} s;
(d) 1.28×10^9 s
21. 10^9 kg
23. (a) 1.083×10^{21} m³;
(b) 5.51×10^3 kg/m³, about 5.5 times the density of water
25. 1.389×10^{-3}
27. 31.29 m/s
29. 2.29 m
31. 5.08 m
33. (a) 30.14; (b) 11740
35. 1.993×10^{-26} kg
37. (a) 1 mi = 1.609 km;
(b) 1 kg = 10^9 μ g;
(c) 1 km/h = 0.278 m/s;
(d) 1 ft³ = 0.0283 m³
39. 7.842×10^3 m/s
41. (a) 0.385 AU; (b) 1.52 AU;
(c) 5.20 AU; (d) 30.1 AU
43. (a) 4; (b) 8
45. (a) 1.283 s; (b) 499 s; (c) 1.50×10^4 s
47. $T: \sqrt{m/k}$
49. $v: \sqrt{gh}$
51. (a) 1; (b) 3; (c) 3; (d) 5
53. 1.50×10^2 cm²
55. 2.700 g/cm³
57. 3×10^9
59. 0.1 mm
61. (a) Earth; (b) $\rho_E/\rho_V = 1.063$
63. (a) 619.54 kg/m³;
(b) The average density is only about 60% of the density of water
65. (a) 16.76 m/s; (b) 1.64
67. about 5×10^{25} atoms
69. (a) 2.563 kg; (b) 4.826×10^{25}

Chapter 2

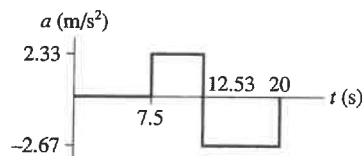
Answers to odd-numbered multiple-choice problems

13. (c)
15. (a)

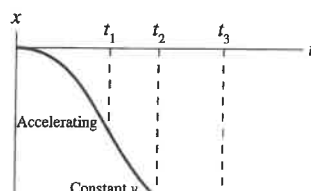
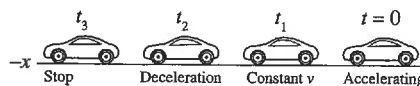
17. (b)
19. (c)
21. (c)
23. (c)
25. (b)

Answers to odd-numbered problems

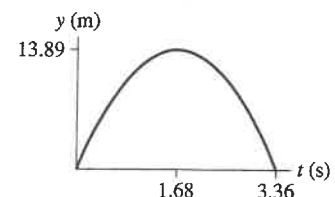
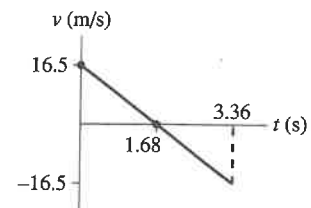
27. $\Delta x = 0$, and s (my notation for total distance) = 400 m
29. (a) $\Delta x = 0$, $s = 960$ km;
(b) $\Delta x = 160$ km, $s = 1120$ km;
(c) $\Delta x = 80$ km, $s = 1200$ km.
31. 500 s or 8 min 20 s
33. 4.44 m/s
35. (a) 204 min = 3.40 h;
(b) 485.3 km/h
37. $\vec{v} = 58$ km/h, $+x$; average speed is 194 km/h
39. 3.96 m/s
41. (a) -3.16 mi/h;
(b) 55.38 s for each mile
43. 2.2 s
45. 0.755c
47. $v(t) = -(55/12) + (5/3)t$
49. (i) 0 – 7.5 s: 2.33 m/s²;
(ii) 7.5 – 12.5 s: 0 m/s²;
(iii) 12.5 – 20 s: -2.67 m/s²



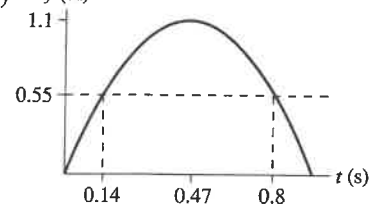
51. (a) a is greatest in the interval 0 – 7.5 s;
(b) a is smallest in the interval 12.5 – 20 s;
(c) a is zero in the interval 7.5 – 12.5 s;
(d) maximum: 2.33 m/s², min: -2.67 m/s²
53. (a) 26.8 m/s; (b) 6.7 m/s²
55. 7.8 m/s
57.



59. (a) -2.41 m/s²;
(b) 5.76 s; yes
61. (a) yes;
(b) 0.332 m/s
63. (a) 25.28 m/s;
(b) 27.27 s;
(c) 386.5 m
65. (a) -9.61×10^5 m/s²;
(b) -9.36×10^5 m/s²
67. 2.014 m/s²
69. -3.71 m/s²
71. (a) 9.58 m/s²;
(b) 37.56 m
73. 13.65 m/s
75. 2.286 s, 22.4 m/s
77.



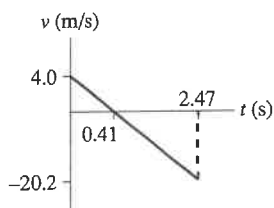
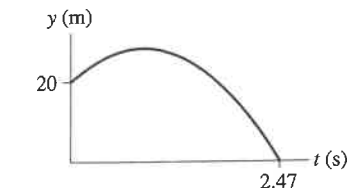
79. 3.60 m/s²
81. (a) 0.948 s;
(b) y (m)



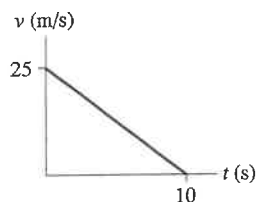
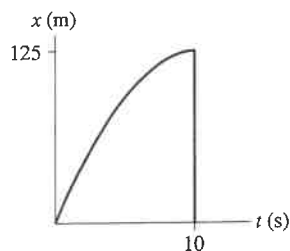
- (c) The player spends about 70% of the time at or above half of the maximum height. This gives the appearance that the player is "hanging" in the air.
83. (a) 138.6 m/s; (b) 1742.4 m;
(c) 184.8 m/s; (d) 44 s
85. -4.95 m/s²

A-12 Answers to Odd-Numbered Exercises and Problems

87. $y = 20 + 4t - 4.9t^2$; $v_y = 4 - 9.8t$



89. (a) 11.2 m/s; (b) -448 m/s^2
 91. (a) -5.99 m/s^2 ; (b) 2.24 s; (c) -12.9 m/s^2
 93. $x = 25t - 1.25t^2$; $v_x = 25 - 2.5t$
 $x(0) = 0$, $x(2) = 45$, $x(4) = 80$,
 $x(6) = 105$, $x(8) = 118.75$,
 $x(10) = 125$ (in meters)



95. (a) 0.554 s, 1.915 s;
 (b) 6.67 m/s, -6.67 m/s
 97. 31.25 m from the left, assuming the moving sidewalk moves to the right.
 99. 90 s
 101. 14.67 s, 161.3 m
 103. 3.57 s; yes, flight time is 3.57 s so the opponent still has time.

Chapter 3

Answers to odd-numbered multiple-choice problems

15. (b)
 17. (c)
 19. (a)
 21. (c)
 23. (a)
 25. (b)

Answers to odd-numbered problems

27. 53.1° , 36.9° , 90°
 29. 1.60 m

31. (a) 15.3 cm;
 (b) 52.3° , 37.6° , 90°
 33. 0.575 km
 35. (a) 117.1° ; (b) 199° ; (c) -73.3°
 37. (i) for $\vec{r}_1$, 5.61 m, -64.76° ;
 (ii) for $\vec{r}_2$, 3.69 m, 164.6°
 39. $(5.95 \text{ m})\hat{i} - (6.05 \text{ m})\hat{j}$
 41. 130.6°
 43. (a) (1.11 m, 6.3 m); (b) (11.26 m, 6.5 m);
 (c) (0, -10 m)
 45. $\vec{T} = \vec{R} + \vec{S} = (10.5 \text{ m})\hat{i} + (7.8 \text{ m})\hat{j}$
 47. $(58.52 \text{ m})\hat{i} - (2.93 \text{ m})\hat{j}$
 49. (a) 28.56 m/s; (b) $-(18.18 \text{ m/s})\hat{i}$;
 (c) $(28.56 \text{ m/s})\hat{j}$
 51. $(9.33 \times 10^{-4} \text{ m/s})\hat{i} +$
 $(7.17 \times 10^{-4} \text{ m/s})\hat{j}$, $1.18 \times 10^{-3} \text{ m/s}$
 53. $(3.54 \text{ m/s}^2)\hat{i} - (8.54 \text{ m/s}^2)\hat{j}$
 55. (a) (245.9 m/s, 917.6 m/s);
 (b) $(4.47 \text{ m/s}^2)\hat{i} + (16.68 \text{ m/s}^2)\hat{j}$
 57. (a) $-(68 \text{ m/s})\hat{i}$;
 (b) $-(9.07 \times 10^4 \text{ m/s}^2)\hat{i}$,
 magnitude: $9.07 \times 10^4 \text{ m/s}^2$, direction: $-\hat{i}$, opposite of the initial direction of the ball's velocity
 59. (a) $(1.14 \text{ m/s}^2, -0.15 \text{ m/s}^2)$;
 (b) $(11.4 \text{ m/s})\hat{i} - (1.5 \text{ m/s})\hat{j}$, speed: 11.5 m/s
 61. (a) $\Delta\vec{v} = +(2.546 \text{ m/s})\hat{j}$
 (b) $\Delta\vec{v}' = +(0.142 \text{ m/s})\hat{i} + (2.403 \text{ m/s})\hat{j}$
 63. (a) 74.4 m; (b) 3.896 s; (c) 18.6 m
 65. (a) 23 m; (b) 1.3 m
 67. (a) 18.66 m/s; (b) 1.392 s;
 (c) $+(13.4 \text{ m/s})\hat{i} - (13.64 \text{ m/s})\hat{j}$
 69. 24.5 m
 71. (a) 383.4 m/s; (b) 55.3 s
 73. 720 m
 75. 9.073 m/s
 77. Between 7.98° and 20.76°
 79. Between 20.4° and 26.57°
 81. 0.0265 m/s^2 ; the ratio is given by $\cos 38^\circ / \cos 0^\circ = 0.788$
 83. (a) At top of the loop, gravity provides the source of centripetal force,
 $mg = mv^2/r$, so $a_r = g$;
 (b) 8.46 m/s
 85. (a) $2.51 \times 10^{-5} \text{ s}$; (b) $7.5 \times 10^{13} \text{ m/s}^2$
 87. 630.6 m/s^2
 89. $4.74 \times 10^5 \text{ m/s}^2$
 91. (d)
 93. (a) 400 m;
 (b) With $A = (0, 0)$, we have
 $\Delta\vec{v} = (80 \text{ m})\hat{i} - (76.4 \text{ m})\hat{j}$
 95. (a) $(21.2 \text{ m/s})\hat{i}$;
 (b) $(21.2 \text{ m/s})\hat{i} - (17.8 \text{ m/s})\hat{j}$;
 (c) 67.2 m
 97. (a) 38.7° , north due east; (b) 0.32 h;
 (c) (10 km, 8.0 km)
 99. (a) 11.2 m/s; (b) 1.62 s
 101. (a) 45° ; (b) 68° ;
 (c) the range is zero because the projectile goes straight up and comes back down

105. $2.72 \times 10^{-3} \text{ m/s}^2$, about $2.78 \times 10^{-4} g$
 107. 0.125 m

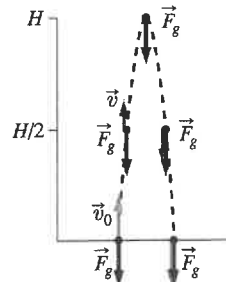
Chapter 4

Answers to odd-numbered multiple-choice problems

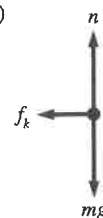
17. (a)
 19. (c)
 21. (c)
 23. (a)
 25. (c)
 27. (c) under the assumption of 25 m/s for the speed of the car.

Answers to odd-numbered problems

29. 2.2 N to the right
 31. 17.43 m/s^2 , upward
 33. 0.184 N, to the left
 35. 0.0698 N, 111.4°
 37. H

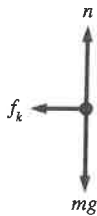


39. $1.64 \times 10^{-21} \text{ m/s}^2$
 41. 0.361 m/s
 43. $7.9 \times 10^5 \text{ N}$
 45. 75 m/s
 47. 15.77 kg
 49. $(-156.1 \text{ N})\hat{i} + (108 \text{ N})\hat{j}$
 51. 31.98 m/s
 53. (a) 477.8 N on hand;
 (b) 19.11 N on head;
 (c) 54.9 N on each leg
 55. 2.97 m/s^2
 57. (a) 291.2 N; (b) 2184 N
 59. 7.82°
 61. 113 N
 63. (a) 12 m/s²;
 (b) 18 N, 12 N, 6.0 N;
 (c) $F_{12} = 18 \text{ N}$, $F_{23} = 6 \text{ N}$
 65. $1.1 \times 10^4 \text{ N}$
 67. (a) 286.7 N; (b) 296.8 N
 69. 2.2 m/s^2
 71. (a) -0.441 m/s^2 ; (b) 6.8 m
 73. (a) -0.0396 m/s^2 ; (b) 37.87 s;
 (c) 0.0040
 75. (a)



- (b) 0.0868

77. (a)


 (b) $mg = 17.15 \text{ N}$, $f = 4.44 \text{ N}$,
 $n = 16.57 \text{ N}$

79. 427.9 m

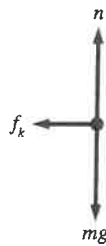
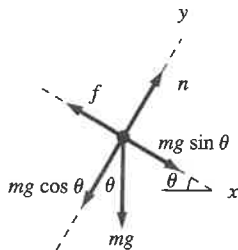
 81. (a) 39.37 m for both car and truck;
 (b) 39.37 m for car going at 50 km/h;
 157.47 m for car going at 100 km/h

83. (a) 116.7 N; (b) 0.594

85. (a) 0.0711;

(b) the system remains at rest

87. (a)

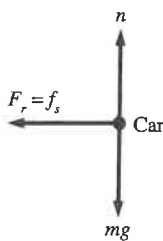

 (b) 7.463 m/s; (c) 47.36 m;
 (d) 23.4 s

 89. (a) $f_s = 966.4 \text{ N} > F_{\max} = 900 \text{ N}$;
 (b) 8.57°

 91. (a) 3.72 m/s^2 ; (b) 2.94 m/s^2

 95. (a) $2.01 \times 10^{20} \text{ N}$; (b) $3.53 \times 10^{22} \text{ N}$;
 $\frac{F_{\text{moon-Earth}}}{F_{\text{Earth-Sun}}} = 5.7 \times 10^{-3}$

97. (a)

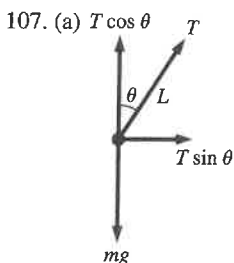


(b) 37.86 m/s

99. 224 days, very close

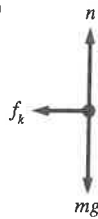
101. (a) 826.5 m; (b) 2 mg

103. (a) 617.4 N; (b) 2498 N; (c) 1263 N

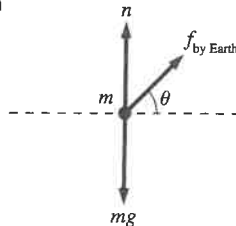
 105. (a) $\frac{mv_{\max}^2}{L} = mg \Rightarrow v_{\max} = \sqrt{Lg}$;
 (b) assuming $L = 1.2 \text{ m}$, $v_{\max} = 3.4 \text{ m/s}$

 (b) $2\pi\sqrt{\frac{L\cos\theta}{g}}$;
 (c) as $\theta \rightarrow 0$, $T \rightarrow 2\pi\sqrt{\frac{L}{g}}$

109. 0.069

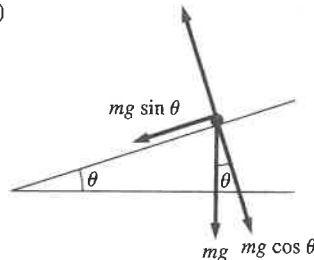
111. (a)


 (b) $2.79 \times 10^4 \text{ N} = 37 \text{ mg}$; (c) 0.788 s

113. (a)

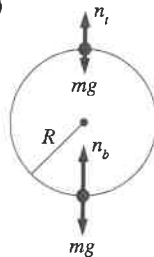

 (b) $(1.484 \times 10^3 \text{ m/s}^2, 2.37 \times 10^3 \text{ m/s}^2)$;
 (c) $F_x = 1.83 \times 10^{-2} \text{ N} = 151.4 \text{ mg}$,
 $F_y = 2.92 \times 10^{-2} \text{ N} = 242.2 \text{ mg}$

115. (a)


 (b) 0.915 m; (c) 1.463 s; (d) 1.25 m/s;
 (e)

117. (a) 0.0073; (b) 186.15 N

119. (a)


 (b) $n_t = 570 \text{ N}$, $n_b = 606 \text{ N}$

Chapter 5

Answers to odd-numbered multiple-choice problems

15. (c)

17. (a)

19. (a)

21. (c)

23. (d)

25. (c)

27. (c)

Answers to odd-numbered problems

29. 1890 J

 31. (a) -2104 N ; (b) $-3.05 \times 10^5 \text{ J}$

 33. -2.59 J

 35. (a) 5.85 J; (b) -5.85 J ; (c) 8.5 J

 37. (a) 4620 J; (b) -1813 J ; (c) 2807 J

 39. (a) 3.17 N; (b) 1.9 J; (c) -1.9 J ;
 (d) 0

 41. (a) 3.92 m/s^2 ;
 (b) 0.294 J on mass 1, and 0.196 J on
 mass 2;

43. (a) 11.76 N/m; (b) 0.833 m

 45. $4.77 \times 10^4 \text{ N/m}$

47. 6.0 J

 49. (a) 1.75 J; (b) 1.3125 J;
 (c) 3.5 J; (d) -1.75 J

 51. (a) -20 J ; (b) 20 J; (c) 80 J;
 (d) 80 J; (e) 20 J

53. 5.39 cm

 55. (a) 0.0392 m; (b) 0.0784 m;
 (c) 0.136 m

57. (a) 3.97 kg; (b) 1220 J; (c) 76.25 J

 59. (a) $1.64 \times 10^8 \text{ J}$; (b) $1.366 \times 10^5 \text{ N}$;
 (c) yes

 61. (a) -15.3 J ; (b) -0.83 N

 63. (a) 34.45 J; (b) -34.45 J ; (c) 25.45 m;
 (d) 1.705 J; (e) 22.33 m/s

65. (a) 14 m/s; (b) 7.5 m

67. 8008 N

 69. (a) $6.5 \times 10^4 \text{ J}$; (b) $7.30 \times 10^4 \text{ J}$

 71. (a) 1.219 m/s; (b) 1.716 m/s;
 (c) 2.211 m/s

 73. $2.58 \times 10^6 \text{ J}$

75. 0

 77. (a) $5.02 \times 10^5 \text{ J}$;
 (b) 0.14 glass

79. about 9300 times; no

81. (a) 32.87 J; (b) 37.84 m/s

83. (a) 5.25 m/s; (b) 69 J

85. 11.38 m/s

87. 0.486 m

89. (a) 6.86 m/s; (b) 1.8 m

 91. (a) $-1.0 \times 10^4 \text{ J}$; (b) $2.1 \times 10^4 \text{ J}$;
 (c) 4.747 m/s

93. 1.014 m

95. 4.915 m/s

97. 2990 N/m

 99. $2.55 \times 10^5 \text{ J}$

 101. $8.983 \times 10^{12} \text{ W}$

 103. (a) 2058 J on person, 603.68 on chair;
 (b) 171.5 W

105. 22.5 m/s

107. 0.617 MJ

 109. (a) 3.822 J;
 (b) a straight line that starts at (0,0) and
 ends at (0.728, 10.5);
 (c) 3.82 J

 111. (a) 95.2 J; (b) 1.586 W;
 (c) Blood is viscous and the passage
 through the blood vessels is much longer
 than the height of the person; some energy
 turns into thermal energy and is not
 recoverable.

113. 1.34 miles

115. (a) 24 J; (b) 12 J

117. (a) 0.288 m; (b) 11.66 N/m

 119. (a) $x = 1 \text{ m}$; (b) $x = 2.917 \text{ m}$

121. (a) 17.96 m; (b) 20.02 m/s

123. -18.29 J
 125. (a) 2.217 m/s ; (b) 1.097 m ;
 (c) 5.33 m/s

Chapter 6

Answers to odd-numbered multiple-choice problems

19. (a)
 21. (b)
 23. (d)
 25. (a)
 27. (d)
 29. (c)
 31. (b)

Answers to odd-numbered problems

33. 196.875 N , $1575 \text{ kg} \cdot \text{m/s}$
 35. $467.2 \text{ kg} \cdot \text{m/s}$
 37. $10.687 \text{ kg} \cdot \text{m/s}$
 39. (a) $1 \text{ N} \cdot \text{s} = 1 (\text{kg} \cdot \text{m/s}^2) \cdot \text{s} = 1 \text{ kg} \cdot \text{m/s}$;
 (b) 53.57 m/s
 41. (a) 0.2156 N ;
 (b) $2.59 \times 10^{-4} \text{ kg} \cdot \text{m/s}$;
 (c) $2.59 \times 10^4 \text{ kg} \cdot \text{m/s}$
 43. (a) $(1.53 \text{ kg} \cdot \text{m/s})\hat{i} + (1.08 \text{ kg} \cdot \text{m/s})\hat{j}$;
 (b) $(2.02 \text{ kg} \cdot \text{m/s})\hat{i} + (1.31 \text{ kg} \cdot \text{m/s})\hat{j}$
 45. (a) $28.95 \text{ kg} \cdot \text{m/s}$, $86.84 \text{ kg} \cdot \text{m/s}$;
 (b) 279.3 J , 837.9 J
 47. (a) $678.4 \text{ kg} \cdot \text{m/s}$; (b) $479.7 \text{ kg} \cdot \text{m/s}$;
 (c) 0
 49. (a) Let $\vec{v}_i = v_i(\cos\theta\hat{i} - \sin\theta\hat{j})$,
 $\vec{v}_f = v_f(\cos\theta\hat{i} + \sin\theta\hat{j})$;
 $\Delta\vec{p} = (0.267 \text{ kg} \cdot \text{m/s})\hat{j}$;
 (b) $(-0.0346 \text{ kg} \cdot \text{m/s})\hat{i} + (0.247 \text{ kg} \cdot \text{m/s})\hat{j}$;
 (c) $(10.68 \text{ N})\hat{j}$ for (a) and $(-1.384 \text{ N})\hat{i} + (9.88 \text{ N})\hat{j}$ for (b)
 51. (a) $1.5 \text{ kg} \cdot \text{m/s}$ for 0 to 0.5 s, and $3.0 \text{ kg} \cdot \text{m/s}$ for 0.5 to 1.0 s;
 (b) 37.5 m/s
 53. (a) $1.46 \text{ kg} \cdot \text{m/s}$; (b) 919.3 kg
 55. 0.477 m/s , in the $-x$ -direction
 57. 17.33 m/s
 59. 1.61 m/s
 61. $-0.186 v_0$ for 60-kg person, and $0.814 v_0$ for the 87.5-kg person (need to know v_0 to have numerical answers)
 65. $v_f = v_i/2$; $K_i = \frac{1}{2}mv_i^2$, and $K_f = \frac{1}{2}(2m)v_f^2 = \frac{1}{4}mv_i^2 = \frac{K_i}{2} \neq K_i$, energy is not conserved
 67. When the bullet hits the block; as the bullet + block system swings to height h .
 69. (a) $mv/(m+M)$; (b) $\frac{M+m}{m}\sqrt{2gh}$
 71. (a) 48.79 n/s ;
 (b) 45.45% ;
 (c) 47.88 m/s
 73. Before: 9.70 m/s ; after: -1.70 m/s
 75. 0.551
 77. 0.05 m/s , and 0.90 m/s

79. $(0.725 \text{ m/s})(\hat{i} + \hat{j})$
 81. 15.4 m/s or 34.5 mi/h , which exceeds 30-mi/h speed limit
 83. One ball 1.167 m/s , 45° above x -axis; other ball 1.167 m/s , 45° below x -axis
 85. $(8.2 \times 10^{-22} \text{ kg} \cdot \text{m/s})\hat{i} + (3.1 \times 10^{-28} \text{ kg} \cdot \text{m/s})\hat{j}$
 87. $(11.5 \text{ m/s})\hat{i} - (23 \text{ m/s})\hat{j}$
 89. 0.40 m
 91. $7.42 \times 10^8 \text{ m}$, outside Sun's radius
 93. 2.063 m from the pivot
 95. (a) 0.30 m/s ;
 (b) velocities are exchanged;
 (c) the CM velocity remains the same
 97. $(x_{\text{cm}}, y_{\text{cm}}) = (54.16 \text{ m}, 28.0 \text{ m})$
 99. (a) 36.39 cm ; (b) 12.55 cm ;
 (c) your muscles contract, pulling the tendons
 103. $(-3.01 \times 10^{-24} \text{ N})\hat{i} - (4.62 \times 10^{-24} \text{ N})\hat{j}$
 105. (a) $(101.8 \text{ N})\hat{i}$; (b) 0.042 m/s
 107. 62.2 cm/s , in the opposite direction
 109. (a) $3.27 \times 10^{-4} \text{ s}$; (b) $2.30 \times 10^4 \text{ N}$;
 (c) 1.875 m/s ; (d) 0.221
 111. 9.6 km/h
 113. (a) 4.276 m/s ; (b) 5.55 m/s
 115. (a) $323 \text{ s} \approx 5.4 \text{ min}$;
 (b) $838 \text{ s} \approx 14 \text{ min}$
 117. (a) $(x_{\text{cm}}, y_{\text{cm}}) = (0, 100.8 \text{ cm})$;
 (b) $(0, 108.9 \text{ cm})$
 119. (a) before: $(0, 98.4 \text{ cm})$, after: $(0, 100.8 \text{ cm})$;
 (b) 2.4 cm
 123. (a) $-v_i/3$; (b) 5

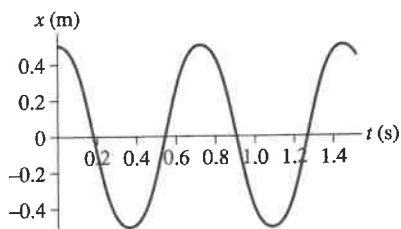
Chapter 7

Answers to odd-numbered multiple-choice problems

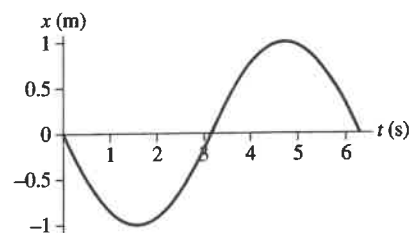
17. (c)
 19. (b)
 21. (b)
 23. (c)
 25. (d)
 27. (a)
 29. (d)

Answers to odd-numbered problems

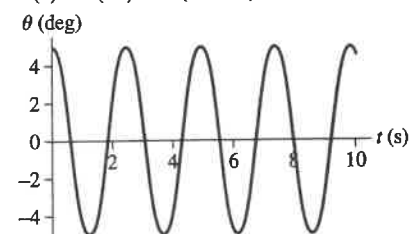
31. (a) periodic; (b) $3.17 \times 10^{-8} \text{ Hz}$
 33. 0.375 s
 35. (a) $6.25 \times 10^{-5} \text{ s}$; (b) $1.01 \times 10^5 \text{ rad/s}$
 37. $T \sim \sqrt{m/k}$
 39. 0.067 N/m
 41. (a) 4.9 m ; (b) 4.62 m ; (c) 3.54 m
 43. 154 N/m
 45. 0.088 kg
 47. $x(t) = (0.5 \text{ m}) \cos(8.71t)$, with $T = 0.721 \text{ s}$



49. (a) $x(t) = A \cos(\omega t + \pi/2)$ (plot with $A = 1$, $\omega = 1$)



51. 0.268 m
 53. (a) 0.733 s ; (b) 0.0187 m
 55. $T/8$
 57. 0.58 m
 59. (a) 14.74 J ; (b) 4.89 m/s ; (c) 0.203
 61. (a) 10 s ; (b) 10.13 kg ;
 (c) $v_{\text{max}} = 0.47 \text{ m/s}$, $a_{\text{max}} = 0.296 \text{ m/s}^2$
 63. (a) 0.295 m ; (b) 0.609 J ; (c) 1.32 m/s
 65. 5.83 Hz
 67. (a) 10 Hz ; (b) 62.83 rad/s
 69. (a) 4.17 m ; (b) no change
 71. (a) 1.28 s ; (b) 0.407 m
 73. $\theta(t) = (5^\circ) \cos(2.556t)$



75. 2.16 s
 77. 0.007 s
 79. lightly damped since ω_{damped} is real and positive

81. (a) $\omega_{\text{damped}} = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2} = 6.778 \text{ rad/s} > 0$;
 (b) $T_{\text{damped}} = 0.927 \text{ s}$, $T_0 = 0.91 \text{ s}$;
 (c) Since $e^{-bT_{\text{damped}}/2m} = 0.3$, the amplitude will damp below $A/2$ in one oscillation
 83. 0.045 kg/s
 85. (a) 196.5 N/m ; (b) 21.6 s
 87. (a) $6.67 \times 10^{-5} \text{ N/m}$;
 (b) $3.446 \times 10^{-10} \text{ kg}$
 89. (a) $T = 0.555 \text{ s}$;
 (b) $T = T_1/\sqrt{2}$, where $T_1 = 0.785$ is the period with one spring
 91. (a) $f_1 = 2f_2 \Rightarrow \omega_1 = 2\omega_2$;
 (b) $a_{1,\text{max}} = 4a_{2,\text{max}}$
 93. (a) 0.144 m ;
 (b) when the normal force on the block is zero (when the acceleration exceeds g)
 95. (a) $v_{\text{max}} = 754 \text{ m/s}$,
 $a_{\text{max}} = 5.68 \times 10^{16} \text{ m/s}^2$;
 (b) $5.70 \times 10^{-21} \text{ J}$
 97. (a) 0.306 m ; (b) $A = 0.222 \text{ m}$;
 $T = 0.583 \text{ s}$

Chapter 8

Answers to odd-numbered multiple-choice problems

15. (d)
17. (a)
19. (b)
21. (a)
23. (d)
25. (c)
27. (b)
29. (a)

Answers to odd-numbered problems

31. $1.267 \times 10^4 \text{ m}$
33. 314 rad/s, about half that of the *E. Coli*'s flagellum.
35. $245 \text{ rad/s} = 2340 \text{ rpm}$
37. $-4.12 \times 10^{-22} \text{ rad/s}^2$
39. 1757.4 turns
41. (a) -119.7 rad/s^2 ; (b) 408 rev; (c) 241 m
43. $-6.13 \times 10^{-22} \text{ rad/s}^2$
45. 5.02 rad/s
47. (a) 97.1 rad/s^2 ; (b) 84.9 rad/s
49. (a) 16.82 m/s ; (b) 0.128 rad/s^2
51. (a) $1.09 \times 10^4 \text{ m}$; (b) 4.74 GB
53. (a) 1.047 rad/s ; (b) 1.1 m/s
55. $2.57 \times 10^{29} \text{ J}$
57. 11 J
59. (a) $7.94 \times 10^{-5} \text{ kg m}^2$;
(b) trans: 35.1 J; rotation: 0.627 J;
 $K_{\text{trans}}/K_{\text{rot}} \approx 56$
61. 40.26 rad/s
65. 102.1 rad/s
67. $\frac{K_{\text{trans}}}{K_{\text{tot}}} = \frac{2}{3}$, $\frac{K_{\text{rot}}}{K_{\text{tot}}} = \frac{1}{3}$
69. 5.42 rad/s
71. $\frac{5}{7} g \sin \theta < g \sin \theta$
73. (a) 0.933 m; (b) the θ dependence cancels out (i.e., the ratio of the times taken to reach the bottom by the two objects does not depend on θ)
75. $22.75 \text{ N} \cdot \text{m}$
77. 33.33 pN
79. (a) $83.3 \text{ N} \cdot \text{m}$; (b) 0.346 rad/s^2
81. 61.67 cm
83. (a) 130.95 N; (b) 194.2 N
85. (a) 0.702 kg; (b) 10.9 N
87. $2.66 \times 10^{40} \text{ kg} \cdot \text{m}^2/\text{s}$, 10^6 times greater than that due to rotation
89. 2.22 rad/s
91. (a) $45.5 \text{ kg} \cdot \text{m}^2/\text{s}$, into the page as viewed from above; (b) $4.55 \text{ N} \cdot \text{m}$, out of the page, as viewed from above
93. $-z$ -direction.
95. $-5.96 \times 10^{16} \text{ N} \cdot \text{m}$, down along the axis of rotation
97. (a) 2.94 Nm; (b) 752.6 rad/s^2
99. (a) 52.5° ; (b) -260.66 J ; (c) $-284.5 \text{ N} \cdot \text{m}$
101. (a) solid sphere wins; (b) solid cylinder: 1.4 m; hollow ball: 1.26 m; hollow cylindrical shell: 1.05 m
103. $1663 \text{ N} \cdot \text{m}$

Chapter 9

Answers to odd-numbered multiple-choice problems

17. (a)
19. (d)
21. (b)
23. (b)
25. (b)
27. (b)

Answers to odd-numbered problems

29. $1.41 \times 10^{-9} \text{ N}$
31. (a) $1.347 \times 10^{26} \text{ N}$;
(b) $F_{\text{Earth-Sun}} = 3.548 \times 10^{22} \text{ N}$,
 $F/F_{\text{Earth-Sun}} = 3.8 \times 10^4$
33. $3.63 \times 10^{-47} \text{ N}$
35. Saturn: 11.1 m/s^2 ; Jupiter: 25.94 m/s^2
37. $6.48 \times 10^{23} \text{ kg}$; very close to the given value of $6.42 \times 10^{23} \text{ kg}$
39. (a) $F_g = 1.27 \times 10^{-10} \text{ N}$;
(b) $6.17 \times 10^{-10} \text{ N} \cdot \text{m}$
41. (a) 3187 m; (b) 32090 m;
(c) $3.446 \times 10^5 \text{ m}$
43. 0.866
45. $2.12 \times 10^{10} \text{ m}$
47. approximately $3.36 \times 10^{18} \text{ m}^3/\text{s}^2$ for Mercury, Venus and Jupiter
49. $T_A/T_B = 2.828$
51. (a) $1.43 \times 10^4 \text{ s}$; (b) $h = 2.175 R_E$
53. 2.732 years
55. (a) $-3.24 \times 10^{35} \text{ J}$;
(b) $-5.27 \times 10^{34} \text{ J}$; (a)/(b) = 6.15
57. (a) 443 m/s; (b) 4120 m/s; (c) 8747 m/s
59. $1.775 \times 10^{32} \text{ J}$; 996 m/s
61. $-1.267 \times 10^{10} \text{ J}$
63. 2375 m/s
65. $6.71 \times 10^7 \text{ m}$
67. 6160 s = 1.7 h
69. (a) 2439 m/s ; (b) $2.1 \times 10^8 \text{ m/s}$
71. $8.84 \times 10^7 \text{ m} = 51 R_{\text{moon}}$
75. (a) $3.8 \times 10^{28} \text{ J}$; (b) $-7.6 \times 10^{28} \text{ J}$;
(c) $-3.8 \times 10^{28} \text{ J}$
77. (a) $4.0 \times 10^{10} \text{ J}$; (b) 1478 kg;
(c) 7360 m/s
79. 7060 s
81. $1.65 \times 10^{12} \text{ m/s}^2$
83. (a) $3.44 \times 10^{-5} \text{ m/s}^2$;
(b) $3.22 \times 10^{-5} \text{ m/s}^2$;
(c) $2.2 \times 10^{-6} \text{ m/s}^2$
85. 0.1796 m/s^2
87. b, d, e, a, c
89. 6.2 m; 22.8 m
91. (a) $4.597 \times 10^{10} \text{ m/s}$; (b) 0.2065
93. (a) 2068 km; (b) 804 km
95. 3.78 m/century
97. 2200 m
99. (a) 2950 m; (b) $2.95 \times 10^{14} \text{ m}$

Chapter 10

Answers to odd-numbered multiple-choice problems

15. (d)
17. (b)
19. (c)
21. (d)
23. (c)

Answers to odd-numbered problems

25. 1.24 m^3
27. 1.69
29. 0.657 L
31. $1.1 \times 10^{-7} \text{ m}$
33. 7.69 kg
35. 0.31 m
37. 10.34 m, not practical
39. (a) $5.81 \times 10^7 \text{ Pa}$; (b) -3.63×10^{-4}
41. 20.1 m
43. 23.8 km
45. 0.898 m
47. $2.29 \times 10^3 \text{ kg}$
49. $1.81 \times 10^6 \text{ N}$
51. $F_b = 0.836 \text{ N}$; $W = mg = 686 \text{ N}$,
 $F_b \ll W$
53. (a) 63700 N; (b) 6.31 m^3 ; (c) 0.904
55. 500 kg/m^3
57. 1033.4 kg/m^3
59. (a) 3.31 N; (b) 2.08 N
61. 17.2%
63. 0.11 m/s^2
65. $2.67 \times 10^{-4} \text{ m}^3/\text{s}$
67. (a) $2.67 \times 10^{-4} \text{ m}^3/\text{s}$; (b) 18 m/s
69. $6.0 \times 10^{-5} \text{ m}^3/\text{s}$
71. 113.6 kPa
73. 3.83 m/s
75. 1800 N, outward
77. decreases by 2.6%
79. 111.4 Pa
81. 0.27 m^2
83. (a) 143 kg; (b) 0.50 m; (c) 7170 Pa
85. $2.28 \times 10^4 \text{ N}$; not likely
87. (a) 1.96 N; (b) $1.33 \times 10^{-2} \text{ m/s}$;
(c) 5.89 m/s
89. (a) 309 m^3 ; (b) 8.4 m
91. $1.5 \times 10^4 \text{ Pa}$
93. (a) 23.7 N; (b) 23.6 N
95. $7.196 \times 10^{-6} \text{ m}^3/\text{s}$

Chapter 11

Answers to odd-numbered multiple-choice problems

15. (d)
17. (b)
19. (b)
21. (b)
23. (b)
25. (d)

Answers to odd-numbered problems

27. (a) 0.566 m/s ; (b) 1.13 m/s
29. 288 km
31. (d) $v = \omega/k$
37. 21.5 cm
39. (a) 235.2 m/s ; (b) $8.86 \times 10^{-4} \text{ kg/m}$
41. G: 392 Hz, 588 Hz; D: 588 Hz, 882 Hz;
A: 880 Hz, 1320 Hz; E: 1318 Hz, 1977 Hz
43. (a) 192.96 m/s
45. 493.8 Hz 4 times every 10 seconds
47. 701.4 m
49. (a) 12005 m; (b) 36.27 s.
51. $2.48 \times 10^{-8} \text{ W}$
53. $\frac{1}{2}$
55. (a) 89.18 dB; (b) 3.16%

57. (a) -6.02 dB; (b) -20 dB; (c) -40 dB
 59. (a) 89.2 km; (b) 892 m
 61. 5.01
 63. 238.15 Hz
 65. 19.94 Hz, 59.83 Hz, 99.71 Hz, 139.6 Hz
 67. (a) 1.53 m; (b) 0.327 m; (c) 0.164 m;
 (d) 0.071 m
 69. 2048 Hz
 71. 6.82 m/s
 73. (a) 247.3 Hz; (b) 238 Hz
 75. 381.4 Hz
 77. (a) 1.2×10^5 Hz, 2.86×10^{-3} m;
 (b) 603 Hz, 0.569 m
 79. 343 m/s
 83. (a) a factor of 4; (b) -6.0 dB
 85. 31.7 dB, louder than a quiet whisper
 at 20 dB
 87. 54.3 kHz
 89. 0.138 m
 91. 0.80 W
 93. 0.0148 m/s

Chapter 12

Answers to odd-numbered multiple-choice problems

15. (d)
 17. (b)
 19. (b)
 21. (c)
 23. (b)

Answers to odd-numbered problems

25. -15°C
 27. (a) -320.8°F , 77.15 K;
 (b) 620.6°F , 600.15 K
 29. 5.4°F
 31. (a) 59.4°F ; (b) -18°C , -0.4°F
 33. 100.76°F
 35. (a) $-40^\circ\text{C} = -40^\circ\text{F}$; (b) 233.15 K
 37. (a) 50.0072 m; (b) 49.982 m
 39. (a) 1.26×10^{-4} m; 4.87×10^{-6} m
 41. -3.12×10^{-9} m
 43. 5.2 s
 47. 0.2532 m²
 49. (a) 39.95 g; (b) 11 g; (c) 52.47 g; (d) 528 g
 51. 3.90×10^{27}
 53. (a) Decrease; (b) 8.43 cm
 55. (a) in order He, Ne, Ar, Kr, Xe, Rn:
 0.164 kg/m³, 0.818 kg/m³, 1.635 kg/m³,
 3.426 kg/m³, 5.37 kg/m³, 9.08 kg/m³;
 (b) only He and Ne
 57. (a) 264.36 kPa; (b) 253.7 kPa
 59. (a) 1.31 g; (b) 1.253 g
 61. 56.14 cm³
 63. (a) less; (b) 1.9×10^4 kg
 65. (a) 1845 m/s; (b) 764.3 m/s
 67. 1172 K
 69. 1.098
 71. (a) 1.2×10^{-19} J; (b) $E/E_{\text{ion}} = 0.055$
 75. -0.138
 77. (a) 5.52×10^{-3} m; (b) 1.392 cm
 79. (a) 19.3633 m; (b) 19.3721 m
 81. 532 kPa
 83. 162.3 atm
 85. $v_{238}/v_{235} = 0.996$
 87. 0.266 m

Chapter 13

Answers to odd-numbered multiple-choice problems

23. (a)
 25. (b)
 27. (a)
 29. (d)
 31. (a)

Answers to odd-numbered problems

33. 1.17×10^6 J
 35. 1.22×10^7 J
 37. 57.5 Cal
 39. 27930 J, or 6.67 Cal
 41. 3.74 kW
 43. (a) 99.2 J/ $^\circ\text{C}$; (b) 19.84 kJ
 45. 19.25°C
 47. 46 g (46 mL)
 49. 427 kg
 51. 268 s ($=4$ min 28 s)
 53. 8841 J/kg
 55. 438 J
 57. 34.5°C , at constant volume
 59. (a) 0.147 kWh; (b) 2.35 cents
 61. 4.62×10^4 J
 63. -5900 J
 65. 266 s
 67. 1.67×10^5 J
 71. 3.0 g
 73. 0.131 kg
 75. 15 W
 77. $\$4.6$
 79. (a) 0.004 m²/ $^\circ\text{C}$ /W;
 (b) 0.0267 m²/ $^\circ\text{C}$ /W;
 (c) 0.133 m²/ $^\circ\text{C}$ /W;
 $R_{\text{Styrofoam}} > R_{\text{wood}} > R_{\text{glass}}$
 81. 0.085 m²/ $^\circ\text{C}$ /W; smaller than R-19 which
 has a value of 3.346 m²/ $^\circ\text{C}$ /W
 83. (a) 1387 W/m²; (b) 3.6×10^6 m²
 85. 0°C , 80 g of ice
 87. (a) ice-water mixture;
 (b) 328.6 g of water and 171.4 g of ice
 89. (a) 1.68×10^5 J; (b) 84 s
 91. 172 s
 93. (a) 4.8×10^4 W (using $k = 40$ W/ $^\circ\text{C}$ m
 for steel, helpful if given in Table 13.4);
 (b) $1.06^\circ\text{C}/\text{s}$ at constant volume
 95. 4.27×10^3 W
 97. 76.8 K
 99. 255 K
 101. 3.14×10^6 s
 103. -25°C
 105. 90 minutes
 107. 0.425°C

Chapter 14

Answers to odd-numbered multiple-choice problems

17. (d)
 19. (a)
 21. (b)
 23. (b)
 25. (c)

Answers to odd-numbered problems

27. 64.86 J
 29. 210 W

31. (a) 415 J; (b) volume decreases;
 (c) -1081.65 J, volume increases
 33. 6810 J; 20430 J
 35. -258 J
 37. (a) 11.7°C , assuming at room tempera-
 ture initially;
 (b) 25.3 J
 39. (a) $4/3$; (b) 215.3 J
 41. (a) 32.42 atm; (b) 258.6 J; (c) 791.7 K
 43. 1565.5 J
 45. (a) 11.21 L; (b) 0.379 atm; 684.5 J
 47. (a) $W_{A \rightarrow B} = -9.0 \times 10^5$ J;
 $W_{B \rightarrow C} = 0$; $W_{C \rightarrow D} = +3.0 \times 10^5$ J;
 $W_{D \rightarrow A} = 0$;
 (b) $W_{\text{net}} = -6.0 \times 10^5$ J;
 49. (a) decreases; (b) 1519.5 J
 51. (a) 25.12 atm; (b) 728.4 K
 53. (a) 288 kJ; (b) 46.9 kJ
 55. (a) 3.36 J/K; (b) -3.36 J/K
 57. Melting: 121.98 J/K; boiling: 605.9 J/K
 59. 2.72 J/K
 61. 254.5 J/K
 63. 0.34
 65. (a) 0.353 ; (b) 600 MJ
 67. 0.268
 69. 1.23×10^{15} J
 71. 2280 W
 73. 0.353
 75. (a) $\$52.7$; (b) $\$17.94$ saved
 81. $C(52, 5) = 2.6 \times 10^6$
 83. (a) 3.3 kJ; (b) 0.392 mol
 85. 1.35
 87. (a) 571.1 kPa; (b) 437.74 J
 89. (a) 0.532 ; (b) 570 MW
 91. (a) 399.4 J; (b) 2.65×10^5 Pa
 93. (a) 288.2 K; (b) 15.5 kJ
 95. (a) 4.29 ; (b) $(\text{COP})_{\text{max}} = 11.42$
 97. (a) 0.635 ; (b) 0.762
 99. (a) 40 kPa; (b) 83.33 kPa; (c) about 80 kJ

Chapter 15

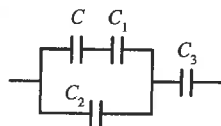
Answers to odd-numbered multiple-choice problems

15. (b)
 17. (b)
 19. (c)
 21. (c)
 23. (d)
 25. (d)

Answers to odd-numbered problems

27. (a) 1.76×10^{11} C; (b) 9.58×10^7 C
 29. (a) 3×10^{26} electrons
 31. 4.8 mC
 33. 0.38 m
 35. (a) 10^{10} N directed straight up; (b) The
 weight is 1.64×10^{-26} N, insignificant
 compared to the electrical force.
 37. 36.4 cm
 39. 5 m. This means that gravity can be
 ignored in atomic or quantum calculations.
 41. $\tan \theta \sin^2 \theta = \frac{kQ^2}{4L^2mg}$

43. (a) Between the charges, 36.6 cm from the smaller charge. (b) Both magnitude and sign of the third charge cancel out of the equations for the Coloumb force when those forces are set equal to each other.
45. 2.77 N directed away from the center of triangle.
47. 8.6×10^{-6} N directed away from the center of the square.
49. (1.78 m, 2.29 m)
51. $(9.92 \times 10^{-9}$ N, -3.29×10^{-8} N)
53. 0.0014 N/C
55. a, d, b, c
57. (a) 8600 N/C in the $+x$ -direction; (b) 3200 N/C in the $-x$ -direction
59. 11°
61. 940,000
63. 8.85×10^{-8} C/m²
65. (a) 5.15×10^{11} N/C; (b) 8.24×10^{-8} N
67. $x = 0.53$ m
69. (a) 23.6 kN/C in the $-x$ -direction; (b) 38.0 kN/C at an angle 71.0° below the $+x$ axis
71. 5500 N/C
73. 185 N/C, 30° below the $-x$ -axis
75. (a) Down; (b) $r = 523$ nm ($d = 1.05$ μ m)
77. 4.13×10^6 m/s, 19.1° below horizontal
79. (a) 9.22×10^6 m/s; (b) 2.54×10^{-15} m. This is on the order of the diameter of a small nucleus.
81. (a) 136 N/C; (b) The same direction as the electron's original motion.
83. 4.6×10^{13}
85. (a) Place the third charge at $y = 2.72$ m; (b) 0
87. (a) 3.33×10^{-12} kg; (b) (i) Up; (ii) Down
89. (a) 3.4×10^8 ; (b) 6.6×10^{-9} N, directed vertically upward; (c) The bee's weight is 1.2×10^{-3} N, roughly 180,000 times the electrical force.
91. $\{+40.3 \mu\text{C}, -6.91 \mu\text{C}\}$ or $\{-40.3 \mu\text{C}, +6.91 \mu\text{C}\}$
93. (a) 2.64 mm; (b) 1.32 mm
95. 4.3 μ m compression
39. 15.82 J
41. (b) 5.25 m/s
43. (a) 140 V; (b) 3.11×10^{-8} C, positive
45. (a) 3.56×10^{-10} C; (b) 2.2×10^9
47. 15.49 m/s
49. 1.15×10^5 V
51. (a) 3.2×10^{-19} C; (b) 3.2×10^{-19} C
53. 2.03×10^3 V/m
55. 1.496×10^6 m/s
57. (a) 3.85×10^4 V/m; (b) -6.15×10^{-15} N
59. -12.65 nC
61. (a) negative; (b) 1.8×10^4 V/m; (c) 4.8×10^{-19} C; (d) 3
63. 2.80 cm
65. (a) 0.70 F; (b) let $C_1 = 0.25$ F, $C_2 = 0.45$ F total $Q: 16.8$ C, $Q_1 = 6.0$ C, $Q_2 = 10.8$ C
67. (a) 17.14 μ C; (b) 8.57 V; (c) $V_1 = V_2 = 3.43$ V; (d) $Q_1 = 3.43 \mu\text{C}$, $Q_2 = 13.72 \mu\text{C}$;
69. 9.0 μ F
71. 1.0 μ F
73. 2.0 μ F
77. (a) 648 J; (b) 0.259 MW
79. (a) 0.443 nF; (b) 22.125 nC; 553 nJ
81. 1250 nF, in parallel
83. (a) 8.87 pF; (b) 106.42 pC; (c) 1920 V
85. 0.0127 mm
87. 2.475 mC
93. (a) 10^5 J; (b) 2.127×10^4 V
97. (a)

(b) 4.27 μ F**Chapter 17****Answers to odd-numbered multiple-choice problems**

19. (b)
21. (c)
23. (c)
25. (c)
27. (d)
29. (b)
31. (a)

Answers to odd-numbered problems

33. 1.29×10^{-5} m/s
35. 8.75×10^{-5} Ω
37. 7.78 kA
39. (a) 8.1 Ω ; (b) 0.081 Ω ; (c) 8.1×10^{-4} Ω
41. Within an uncertainty of $\Delta R = 0.0228$ Ω
43. 4
45. (a) 125 s; (b) 12 J
47. 4.48 Ω
49. (a) 3.6 C; (b) 5040 C; (c) 6048 J
51. 3.6 V, 1.94 Ω
53. (a) 149 Ω ; (b) 0.048 A (in 250- Ω resistor) and 0.032 A (in 370- Ω resistor)

59. 40 Ω
61. 75 V
63. (a) 2.5 V; (b) let $(R_1, R_2, R_3) = (20 \text{ k}\Omega, 30 \text{ k}\Omega, 75 \text{ k}\Omega)$, $(I_1, I_2, I_3) = (0.125 \text{ mA}, 0.0833 \text{ mA}, 0.0333 \text{ mA})$

65. 230.4 J
67. 0.042 A
69. (a) 14.58 A; (b) 1.05 MJ = 0.292 kWh; (c) \$1.31
71. 12
73. (a) 870 W; (b) 1.81°C
75. (a) 0.034 nC; (b) 0.316 nC; (c) 1.69 nC
77. (a) 0.693; (b) 1.228
79. (a) 2.4 mC; (b) 0.635 mA
81. (a) 0.48 C; (b) 0.0229 C more
83. (a) 5.735 mC, 5.479 mA; (b) 39.4 mC, 2.417 mA
87. 20.8 Ω , 1.84 V
89. 3.367 μ F
91. (a) 6800 Ω ; (b) 1.632 s
95. (a) 4800; (b) 1.0 A

Chapter 18**Answers to odd-numbered multiple-choice problems**

23. (b)
25. (d)
27. (d)
29. (c)
31. (a)
33. (d)

Answers to odd-numbered problems

35. (a) 0.1995 μ N, down; (b) 0.1995 μ N, up
37. (a) 5.30 N, vertically downward; (b) 5.30 N, vertically upward; (c) 5.30 N, vertically upward; (d) 5.30 N, vertically downward; (e) 0
39. (a) 1.50×10^{-18} N, west; (b) 9.0×10^8 m/s², west
41. (a) 0.1088 pN, $-z$ -direction; (b) 6.8×10^5 N/C, $-z$ -direction
43. (2025 N/C), $-x$ -direction
45. 1.87×10^{-28} kg; $m_\mu/m_p = 0.112$, $m_\mu/m_e = 205$
47. (a) 1.885×10^5 m/s; (b) 0.052 T
49. 0.69 mT
51. (a) 85.37 μ m; (b) 3.655 mm; (c) 3.644 mm
53. 0.0876 T
55. 38.59 mA
57. (a) 54208 T; (b) 54.208 T
59. assuming current flows eastward, $B = 73.3 \mu\text{T}$, north
61. (a) 0; (b) $-(0.09 \text{ N} \cdot \text{m})\hat{j}$; (c) $(0.0636 \text{ N} \cdot \text{m})(\hat{i} - \hat{j})$
63. $(0.01875 \text{ N})\hat{j}$ on segment along x , $(0.01875 \text{ N})\hat{i}$ on segment along y , and $-(0.01875 \text{ N})(\hat{i} + \hat{j})$ along diagonal; the net force is zero

Chapter 16**Answers to odd-numbered multiple-choice problems**

19. (b)
21. (d)
23. (c)
25. (b)
27. (d)
29. (c)
31. (c)

Answers to odd-numbered problems

33. 3.15×10^{14} J
35. (a) earth: 4.4×10^4 C, moon: 546 C; (b) earth: 8.7×10^{11} C/m², moon: 1.4×10^{11} C/m²
37. -1.99×10^7 J

65. (a) $70.5 \mu\text{N}$, west;
(b) $75 \mu\text{N}$, 70° north of up;
67. $11.04 \text{ A} \cdot \text{m}^2$
69. $R = 1.99 \text{ cm}$, 12 turns
71. 0.0768 T
73. $2.92 \mu\text{T}$
75. (a) $133.3 \mu\text{T}$; (b) $16.67 \mu\text{T}$;
(c) 83.33 N/m , repulsive
77. $3.35 \times 10^{-3} \text{ N/m}$, toward each other,
force attractive
79. (a) 0;
(b) $1.326 \times 10^{-4} \text{ N/m}$, toward the center
of square
81. (a) $2.356 \times 10^{-4} \text{ T}$; (b) 0
83. (a) $(8.6 \times 10^{-7} \text{ N})$, toward the wire;
(b) $(8.6 \times 10^{-7} \text{ N})$, toward the loop;
(c) $(8.6 \times 10^{-7} \text{ N})$, away from the wire;
(d) $(8.6 \times 10^{-7} \text{ N})$, away from the loop
85. (a) 0.102 T ; (b) 0.02356 T ;
(c) 0.0628 T in both cases
87. $-(147.6 \text{ m/s})\hat{i}$
89. $(0.06 \text{ N})\hat{i}$
91. 0.726 T
93. 22.62° , west of magnetic north
95. 1000 m

Chapter 19

Answers to odd-numbered multiple-choice problems

17. (a)
19. (c)
21. (a)
23. (a)
25. (b)
26. (c)

Answers to odd-numbered problems

27. 0.0736 Wb
29. 0.0117 Wb
31. 0.2873 A
33. (a) Clockwise;
(b) 0.6136 mA ; 0.3375 mV
35. 35
37. (a) 0.0751 , clockwise; (b) 12.98 mW
39. (a) 1.764 V ; (b) 3.528 V
41. (b) 1.225 V
45. (a) 0.01178 V ; (b) 1508 V
47. 50
49. (a) 17.5Ω ;
(b) All power would be lost.
51. 0.06 s
53. (a) 11.5 mH ; (b) 9.325 A ;
55. (a) 4.8 A ; (b) 4.15 A ; (c) 0.03475 s ;
(d) 4.15 A
57. Between 2516 and 3559 Hz
59. Increase C fourfold
61. (a) 0.0393 s ; (b) 0.0731 J ;
(c) 8.55 mC
63. (a) 0.0687 A ; (b) 16.51 W
65. (a) 280 V ; (b) 560 W
67. (a) 169.7 V ; (b) 49 Hz ;
(c) $\mathcal{E}(t) = (169.7 \text{ V}) \sin(308t)$
69. (a) 31.83 Hz ; (b) 79.58 Hz , 150Ω
71. 129.5Ω
73. (a) 0.33929 A ; (b) 0.33938 A

75. (a) Capacitive reactance: 1061Ω , inductive reactance: 84.82Ω ;
(b) 1666Ω ; (c) $31.27 \mu\text{F}$
77. (a) -27.7° ; (b) 0.885 ; (c) 75.3 W
79. (a) 516.4 rad/s ; (b) 10.87 W
81. 42
83. 1102.9 V
85. (a) 3.573 A/s ; (b) Opposite
87. (a) $5.07 \times 10^{16} \text{ J}$;
(b) 3380 s , or 56.3 minutes

Chapter 20

Answers to odd-numbered multiple-choice problems

13. (d)
15. (d)
17. (b)
19. (d)
21. (d)
23. (a)

Answers to odd-numbered problems

25. $4.74 \times 10^{14} \text{ Hz}$
27. 273 nm
29. 468 nm
31. (a) $0.50 \mu\text{T}$; (b) 7.5 N/C
33. (a) $1.18 \times 10^{-4} \text{ Pa}$; (b) $5.9 \times 10^{-5} \text{ Pa}$
35. (a) 0.15 MHz , radio waves;
(b) 150 MHz , radio waves;
(c) 150 GHz , microwaves;
(d) $1.5 \times 10^{14} \text{ Hz}$, IR;
(e) $1.5 \times 10^{17} \text{ Hz}$, UV
37. $10^9 - 10^{12} \text{ Hz}$
39. 1.50 m
41. $3 \times 10^8 \text{ m/s}$
43. $t_{\text{perp}} = 35.36 \text{ s}$; $t_{\text{perp}} < t_{\text{parallel}}$
45. (a) Earth: 133.3 s , ship: 133.3267 s ;
(b) Earth: 4.44 s , ship: 4.24 s ;
(c) Earth: 1.778 s , ship: 1.176 s ;
(d) Earth: 1.3468 s , ship: 0.190 s ;
47. (a) 10 ns ; (b) 11.547 ns ;
(c) 3.464 m ; (d) $3 \times 10^8 \text{ m/s}$
49. 0.141 c
51. 59.945 s
53. 0.9999995 c
55. 709 m
57. (a) Yes, $771.442 \text{ m} < 75 \text{ m}$; (b) 0.033 s
59. 0.9803 c
61. 0.9396 c
63. (a) 534.6 nm ; (b) 488.45 nm ;
(c) 311.77 nm ; (d) 123.88 nm ;
(e) 38.28 nm
65. 0.47 c , to the right
67. $f_0 \sqrt{\frac{1+v/c}{1-v/c}}$; $2.29 \times 10^{15} \text{ Hz}$
69. $34.4/\text{min}$
71. 0.999949 c
75. 1.022 MeV
77. 0.866 c
79. (a) 0.14 c ; (b) 0.910527 c
81. (a) $2.733 \times 10^{-20} \text{ kg} \cdot \text{m/s}$,
 $E = 51.1 \text{ MeV}$; (c) 100
83. 0.995 c
85. 7.94 MW
87. $1.067 \times 10^{-16} \text{ kg}$

89. 0.96 c
91. 0.93237 c

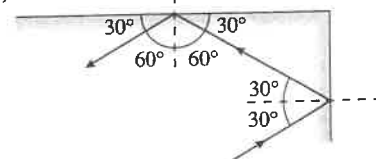
Chapter 21

Answers to odd-numbered multiple-choice problems

17. (b)
19. (c)
21. (a)
23. (c)
25. (b)

Answers to odd-numbered problems

27. 6 m
29. (a) 4.0 m ; (b) 0.50 m/s
31. (a)



- (b) 24 cm
33. (a) 116.7 cm ; (b) 53.85 cm
39. (a) 187.5 cm ; (b) real; (c) -1.5
41. $d_i = 52.2 \text{ cm}$; $M = -1.5$, image inverted
45. (a) Upright;
(b) 2.25 cm from center of the ball;
(c) 1.5 cm
47. (a) 40 cm ;
(b) image located at $d_i = -15 \text{ cm}$,
upright, reduced with $M = 0.25$
49. Diamond with $n = 2.42$, $v = 0.413 \text{ c} =$
 $1.28 \times 10^8 \text{ m/s}$
51. (a) $2.29 \times 10^8 \text{ m/s}$; (b) $1.948 \times 10^8 \text{ m/s}$;
(c) $1.27 \times 10^8 \text{ m/s}$; (d) $1.24 \times 10^8 \text{ m/s}$
53. (a) 14.9° (b) 12.92°
55. 0.55 m
57. 0.93 m
59. $\theta_c = 24.4^\circ$
61. (a) 22.48° ; (b) 0.469 mm
67. (a) $d_o = 60 \text{ cm}$, $d_i = 30 \text{ cm}$;
(b) $d_o = 30 \text{ cm}$, $d_i = 60 \text{ cm}$
59. (a) Convex; (b) Inverted; (c) $d_o = 15 \text{ cm}$
71. (a) 4.91 mm ; (b) 9.0 mm ; (c) 54 mm
73. 4.89 cm
75. (a) 6.56 ; (b) 17.67
77. (a) 0.172° ; (b) 1.72°
79. 1.87 cm
85. (a) 23.0 cm ; (b) Switching the two values
with $R_1 = 20 \text{ cm}$, and $R_2 = -40 \text{ cm}$
gives the same result.
87. -2.67 diopters
89. 2.03 diopters , assuming glasses 2 cm
from eyes
91. (a) $f = \frac{R}{n-1}$; (b) 13 cm
93. $n_2 = 1 + \frac{1}{3}(n_1 - 1)$
95. (a) 19.95 cm ;
(b) $M = -0.5536$, image size reduced
97. (a) 72 cm ;
(b) $d_i = -4.5 \text{ cm}$, or 4.5 cm inside the
surface

99. (i) for $M = -1/2$: (a) $d_o = 56$ cm,
 $d_i = 28$ cm; (b) image real;
 (c) $f = 18.67$ cm;
 (ii) for $M = -2$: (a) $d_o = 28$ cm,
 $d_i = 56$ cm; (b) image real;
 (c) $f = 18.67$ cm;
 101. converging lens with $P = +3.33$ diopters
 103. 7.56 mm

Chapter 22

Answers to odd-numbered multiple-choice problems

15. (d)
 17. (d)
 19. (b)
 21. (b)
 23. (c)

Answers to odd-numbered problems

25. (a) 76.16 nm; (b) 96.03 nm;
 (c) 105.96 nm
 27. (a) No violet light;
 (b) Violet light
 29. 542.64 nm, green light
 31. 176.5 nm
 33. 2945 nm
 35. (a) 2.10 mm; (b) 3.155 mm;
 37. (a) 0.364° , 0.728° and 1.092° ; (b) Yes
 39. 526.45 nm
 41. 25
 43. 11.81 mm
 45. 5.33 μm , with $\lambda = 1.0$ nm
 47. (a) 550 nm; (b) yes, 61.63°
 49. (a) 12.37° ; (b) 23.13° ; (c) 40°
 51. (a) Entire visible spectrum;
 (b) $500 - 700$ nm (green to violet);
 (c) No part of visible spectrum
 53. (a) 0.02967° ; (b) 0.1253°
 55. (a) 68.76 μm ; (b) 99.3 μm
 57. 0.812 mm, using $\sin \theta_2 = 2.23 \frac{\lambda}{D}$
 59. 1.82 cm
 61. (a) 2.796×10^{-7} rad;
 (b) 2.645×10^{10} m, or about $0.176 d_{SE}$,
 the Earth-Sun distance
 63. 549.7 nm
 65. 0.67
 67. (a) 5.0 mW; (b) 3.75 mW
 69. (a) 0.1875 ; (b) 0.1875 ; (c) 3.045×10^{-4}
 71. (a) 0.25 ; (b) 0
 73. 1.34 m, with $\lambda = 550$ nm
 75. 0.33 m
 77. 1.98 s
 79. 0.079
 81. 210 nm

Chapter 23

Answers to odd-numbered multiple-choice problems

17. (b)
 19. (c)
 21. (d)
 23. (d)
 25. (d)
 27. (d)

Answers to odd-numbered problems

29. 6.25×10^{18}
 31. (a) 139.6 g;
 (b) 2.4×10^{-7} C, too small to be noticed
 33. 2282 K
 35. 7431 K
 37. (a) 1369 nm, infrared;
 (b) 2.18×10^{-5} m²
 39. 2.73 K
 41. 951 K
 43. 1.028×10^{15} Hz, 292 nm
 45. (a) 1.12×10^{15} Hz, 267.5 nm; (b) 4.56 V
 47. (a) 0.448 V; (b) 3.97×10^5 m/s
 49. (a) 2.32 V; (b) 3.926×10^{-19} J
 51. (a) 6.272×10^{-34} J $\cdot$ s, -5.34% ;
 (b) 7.2×10^{14} Hz, 4.5×10^{-19} J
 53. (a) 6.626×10^{-25} J;
 (b) 1.988×10^{-22} J
 55. (a) 8.28×10^{-20} J; (b) 1.21×10^{20}
 57. (a) UVB: 1.56×10^{18} , visible:
 2.77×10^{18}
 59. (a) 4.53×10^{-10} ; (b) 1387 W/m²
 61. 90°
 63. 108.5°
 65. 97.2 pm
 67. (a) 1876.5 MeV;
 (b) Each photon has an energy of
 938.27 MeV
 69. (a) 7.27×10^5 m/s; (b) 546 fm
 71. 2.91×10^6 m/s
 73. (a) 16.73 eV; (b) 1.46×10^{-21} J;
 (c) 3.65×10^{-22} J
 75. 27.82 pm, very unlikely
 77. (a) 1.506×10^{-24} J;
 (b) 4.96×10^{-19} J
 79. (a) 5.273×10^{-25} kg $\cdot$ m/s;
 (b) 1.526×10^{-19} J; (c) 1.3 μm
 81. (a) 24.1 pm; (b) 4.2 pm; (c) 0.345 pm
 83. 0.02478 fm, much smaller than the diam-
 eter of a proton
 85. (a) 1.32 fm; (b) 9.3827×10^8 eV
 87. 1.988×10^{-26} m
 89. 294 eV
 91. (a) 2.77×10^{20} ; (b) 6.1×10^9

Chapter 24

Answers to odd-numbered multiple-choice problems

17. (c)
 19. (a)
 21. (d)
 23. (b)
 25. (d)

Answers to odd-numbered problems

27. 1.92×10^7 m/s
 29. 4.86 fm with Al target, much shorter than
 29.6 fm found in Ex. 24.1 for Au target
 31. (a) Atom: 6.236×10^{-31} m³, nucleus:
 7.238×10^{-45} m³;
 (b) 2.68×10^3 kg/m³, about 2.7 times
 the density of water;
 (c) 2.31×10^{17} kg/m³, much greater than
 that found in atom for part (b)
 33. (a) 13; (b) 28; (c) 41;

35. $2279 - 7458$ nm
 37. (a) 3.4 eV; (b) 365 nm
 39. (a) 2.19×10^6 m/s; (b) 1.09×10^6 m/s
 41. (a) 486 nm; (b) 0.82 m/s
 43. (a) 1.89 eV; (b) absorption; (c) 656 nm
 45. 69
 49. (a) -54.4 eV; (b) -122.4 eV
 51. (a) $2s$; (b) $4p$; (c) $4f$; (d) $3d$
 53. (a) Allowed, 656 nm;
 (b) Forbidden;
 (c) Allowed, 1875 nm;
 (d) Allowed, 102.6 nm
 55. (a) $l = 1$;
 (b) $m_l = -1, 0, +1$;
 (c) $m_s = \pm \frac{1}{2}$
 57. (a) $\frac{L_f}{L_i} = \frac{1}{\sqrt{3}}$; (b) $\frac{L_f}{L_i} = \sqrt{3}$;
 (c) $\frac{L_f}{L_i} = \frac{1}{\sqrt{2}}$
 63. (a) Holes; (b) Electrons
 65. 40 kV
 67. 0.0248 fm
 69. 6895 V
 71. 2.33 eV
 73. (a) 532 nm; (b) 1.68 W
 75. $n = 5 \rightarrow 3$
 79. (a) 35.3° , 65.9° , 90° , 114.1° , 144.7° ;
 (b) $L_z = m_l \frac{h}{2\pi}$, $m_l = 2, 1, 0, -1, -2$

81. 3.02×10^{17} per second

Chapter 25

Answers to odd-numbered multiple-choice problems

17. (a)
 19. (b)
 12. (d)
 23. (b)
 25. (b)
 27. (c)

Answers to odd-numbered problems

29. (a) ^{49}Cr ; (b) ^{94}Tc ; (c) ^{190}Pb ;
 31. For ^{80}Br : $Z = 35$, $N = 45$;
 ^{80}Kr : $Z = 36$, $N = 44$;
 33. (a) 2.3 fm; (b) 3.26 fm;
 (c) 6.13 fm; (d) 7.45 fm
 35. Any nuclide with $A = 216$ (e.g., ^{216}At)
 45. (a) 2.66×10^{33} J; (b) 2.66×10^{33} J;
 (c) 2.93×10^{16} kg
 47. (a) $^{67}\text{Cu} \rightarrow \beta^- + ^{67}\text{Zn}$;
 (b) $^{85}\text{Kr} \rightarrow \beta^- + ^{85}\text{Rb}$;
 (c) $^{112}\text{Pd} \rightarrow \beta^- + ^{112}\text{Ag}$
 49. (a) ^{158}Er ; (b) β^+ ; (c) ^4He ; (d) ^4He
 51. $^{210}\text{Bi} \rightarrow \beta^- + ^{210}\text{Po}$, 1.16 MeV
 53. (a) 0.148 Bq/L;
 (b) 1.28×10^4 /L, assuming constant rate
 55. Electron: 0.45 mm; alpha: 19.3 mm
 57. (a) 0.325 mg; (b) 0.081 mg
 59. 9.32 mg
 61. (a) 1.42×10^{11} Bq;
 (b) 6.16×10^8 Bq, with 0.108 μg
 remained

63. 8.92 kg
 65. About 6.3×10^9 years
 67. (a) 3.11 MeV released;
 (b) $Q = -10.4$ MeV, $K_{\min} = 11.9$ MeV;
 (c) 2.21 MeV released
 69. (a) $Q = -4.3$ MeV;
 (b) $Q = -2.92$ MeV;
 (c) $Q = -6.09$ MeV (I simply give the Q value, and not $K_{\min}$)
 71. (a) ^{89}Kr ; (b) ^{143}La ;
 (c) ^{96}Zr
 73. (a) ^{239}Np ; (b) $2n$; (c) ^{120}Ag
 75. 3.5 kg/day
 77. 0.776 kg, much smaller than the 50 kg used in early bomb
 79. (a) $Q = -22.37$ MeV;
 (b) $Q = -7.46$ MeV;
 (c) 17.6 MeV
 81. (a) 12.86 MeV; (b) 17.35 MeV;
 (c) 18.35 MeV
 83. 0.0307 mol each (0.09 g of ^3H and 0.06g of ^2H)
 85. 11.45 MeV, 0.11 pm
 87. (a) 2.5×10^{-31} kg; (b) 9.74×10^{10} Bq;
 (c) After 1 week: 1.92×10^{-15} g; after 30 days: 4.32×10^{-43} g
 89. Flight: 2.5×10^{-6} , or 1 in 400,000; PET scan: 5×10^{-4} , or 1 in 2,000
 91. $^{198}\text{Hg} + ^1_0n \rightarrow ^{197}\text{Ag} + ^2_1\text{H}$
 93. Yes; takes only 4.29 days to drop to 2000 Bq/L

Chapter 26

Answers to odd-numbered multiple-choice problems

13. (c)
 15. (c)
 17. (b)
 19. (c)
 21. (c)

Answers to odd-numbered problems

23. 0.511 MeV, 2.43 pm
 25. 0.866c
 27. 1250 kg total (625 kg of matter and 625 kg of anti-matter)
 29. 1.32×10^{-18} m
 31. 0.252 fm
 33. (a) 1.24 TeV;
 (b) 1.24 TeV
 35. (a) $\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$;
 (b) $\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$
 37. 105.66 MeV, 11.7 fm
 39. (a) $\bar{\nu}_e + p \rightarrow n + e^+$;
 (b) $K^+ \rightarrow \mu^+ + \nu_\mu$
 41. (a) Allowed;
 (b) Not allowed, violation of baryon number conservation and muon lepton number conservation;
 (c) Not allowed, electric charge not conserved

45. (a) 62 MeV;
 (b) K^{-1} gets more
 47. $\bar{s}\bar{s}\bar{s}$
 49. (a) $\bar{c}u$;
 (b) Same mass and lifetime, different charm number and decay products
 55. (a) uuu, ccc, ttt ; (b) $\bar{u}\bar{u}\bar{u}, \bar{c}\bar{c}\bar{c}, \bar{t}\bar{t}\bar{t}$
 57. (a) $v_e = 0.999\,999\,87$ c;
 (b) $v_p = 0.875$ c, $v_e/v_p = 1.143$
 59. (a) $2.668 \mu\text{s}$
 61. (a) Colliding beam: 2.0 GeV, with stationary target: 2.32 GeV;
 (b) Colliding beam: 20 GeV, with stationary target: 4.72 GeV;
 (c) Colliding beam: 200 GeV, with stationary target: 13.82 GeV
 63. 5.42×10^{16} K
 65. 2.73 K
 67. 2.87×10^8 m/s
 69. (a) 1.29 MeV;
 (b) 9.98×10^9 K; $< 10^{-3}$ s after Big Bang
 71. No, baryon number not conserved
 73. 7.0×10^{19} m
 75. (a) 7462;
 (b) $v = c(1 - 8.98 \times 10^{-9})$
 77. (a) 2.556 fm; (b) -2.815 keV